

A review of fairness challenges in natural language processing

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ABSTRACT

Natural Language Processing (NLP) systems are increasingly deployed in high-stakes systems including healthcare, education, recruitment, and law enforcement, yet they have frequently coded and magnified biases that undercut their system's fairness and trust. This review synthesizes and critically analyzes 121 studies that were published in the year 2014 and up to date that address bias in NLP. We present a novel taxonomy of 18 bias types, such as previously underexplored categories like geographic, disability, and annotation bias, and project them onto the NLP lifecycle, taking data as the starting point to deployment. Four key detection paradigms are examined (statistical, model-probing, benchmark-based, and human-centric), alongside mitigation strategies at the data, model, and post-processing levels. Unlike prior surveys, this study offers a lifecycle-aware framework that connects bias origins, detection methods, and mitigation practices, while focusing on persistent challenges such as intersectionality, generalization, and fairness-performance trade-offs in large language models (LLMs). We argue that achieving fairness in NLP requires not only technical interventions but also socio-technical approaches that integrate community participation, transparency, and governance. Offering a structured, critical, and forward-looking synthesis, this work contributes a roadmap for building transparent, equitable, and socially responsible NLP systems.

1. Introduction

Natural Language Processing (NLP) technologies have revolutionized the interaction between humans and machines, enabling applications ranging from virtual assistants and search engines to automated hiring systems and content-moderation algorithms. These technologies now impact major decisions on the lives of millions of individuals on a daily basis, from who receives a job to what content we see in our social media feeds. But beneath the neutral-looking surface of these AI-driven systems is an inherent difficulty: bias.

Bias in NLP refers to systematic unfairness in language models. Unlike human bias, which may be conscious or unconscious, algorithmic bias operates at scale, amplifying discrimination across entire populations. Suppose a Natural Language Processing (NLP) system is biased. In that case, it does not just render one unfair decision- it will probably replicate this unfair decision across thousands or millions of interactions, disadvantaging entire communities systematically.

The common ground phenomenon of biased predictive models in NLP is not new, as the community has long worked on the domain

adaptation problem, where models trained on a specific type of data perform better than others, whereas others perform poorly [1]. This problem arises from the tendency of statistical analytical models to pick up non-generalizable signals during the training process [1].

A huge part of the researchers are now focused on this 'bias' in Natural Language Processing (NLP), including works on 'bias' in embedding spaces, since it is a major influencer in a system that moderates many things in this era [2,3] such as language modeling, coreference resolution, and machine translation [4]. These biases in large language models (LLMs) have the possibility of originating directly from the data selection process, a rarity termed a "data selection bias", which can lead to an unbalanced representation of different cultures if the training corpus has a skewed distribution of languages [5]. When an NLP system exhibits bias, it does not just make isolated mistakes - it risks replicating and reinforcing societal inequalities on an unprecedented scale. The roots of this problem are deep and stem from the way language models are trained and evaluated. These biases have far-reaching effects that go

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Table 1
Summary of detection metrics and their applications.

Metrics name	What it measures	Typical application context	Example bias types	Ref.
WEAT/SEAT	Strength of association between concepts and attributes in embedding spaces.	Analysis of static and contextualized word/sentence embeddings.	Racial, Gender, Age	[10,11]
p-% Rule/Disparate impact ratio	Ratio of positive outcome rates between different demographic groups.	Fairness auditing of binary classification models (e.g., hiring).	Racial, Gender, Economic	[12,13]
Equalized Odds	Whether a model has similar true positive and false positive rates across groups.	Predictive models where error fairness is critical (e.g., healthcare).	Racial, Model, Deployment	[14,15]
Inter-Annotator Agreement (IAA)	Consistency between human annotators.	Quantifying subjectivity and reliability in labeled training data.	Annotation bias	[16,17]
Median Error	Spatial accuracy in geolocation tasks,	Evaluating performance of geographic parsing systems.	Geographic bias	[18]
Distance (MdnED)	highlighting regional disparities.			

far beyond technical limitations and raise ethical and societal concerns. The main issues of this problem are deeply embedded in how language models are trained and evaluated [6]. Therefore, understanding and mitigating these biases are important for developing equitable and trustworthy NLP technologies. This study provides a comprehensive review of biases in NLP systems by systematically categorizing and examining a wide range of major and minor biases, their underlying causes, and their implications for fairness and equity in NLP applications. This field has seen a significant increase in research on bias, with a focus on understanding both the origins of bias and developing methods for detection and mitigation [7]. However, the study of bias in NLP systems often lacks coherent moral reasoning, making it difficult to identify which actions of the system are harmful, to whom, and why [6]. This barrier demonstrates the urgent need for a more critical approach that examines the power dynamics between technologists and the people affected by these systems and focuses on their lived experiences [6].

Recent work on predictive biases in NLP models provides a unifying framework that differentiates between outcome and error inequalities [8]. This framework identifies key origins of bias, such as label bias, selection bias, model bias, and semantic bias [8]. Semantic bias arises from issues within embedding spaces, which are crucial for many NLP tasks [8]. The impact of biases extends across various NLP applications, including language modeling, coreference resolution, machine translation, sentiment analysis, and hate speech/toxicity detection [6].

A critical aspect of addressing bias is the actionability of bias measures. Researchers are increasingly questioning whether existing metrics provide sufficient information to enable informed action for bias mitigation [4]. Many current bias evaluation metrics are often unclear or lack details on their intended use and reliability assessment, creating a significant gap in the development and reporting of bias measures [4]. This calls for a change to policies that explicitly direct mitigation and intervention efforts, connecting bias detection to particular consequences. [9]. A fundamental step in this process is understanding the available tools. Table 1 provides a summary of common detection metrics and their specific applications, highlighting the connection between measurement and actionable insight.

Mitigation strategies often involve changing the model architecture or training process [5]. An effective method is “self-diagnosis and self-debiasing”, where pre-trained language models can identify their own biases and reduce the generation of problematic text [19]. However, fully eliminating bias remains a challenge for large language models due to their immense size and limited interpretability [7]. Another approach involves using Semantic Web technologies to integrate formal knowledge representations, which can lead to fairer AI systems [20]. These technologies provide structured contextual knowledge to reduce reliance on biased patterns in raw data and address issues like data drifts and sparsity [20].

Ultimately, addressing bias in NLP requires continuous research and collaboration among computer scientists, social scientists, and ethicists to develop equitable and responsible technologies [6]. Although there are 18 different types of bias in the taxonomy, to grasp the significance of each bias from a strategic standpoint, one must assess essential

factors that go past classification. It becomes apparent that essential mismatches appear in the relevance of research versus application. It emerges that Gender bias and Racial bias, thoroughly explored topics (together 16% of literature analysis) are not the most significantly affected groups. On the contrary, both Geographic bias, one of the least explored biases (only 5% literature contribution) that affects 4+ billion Global South users, Disability bias with 1.6 billion affected people, again with limited literature contribution, or Confirmation bias, one of the least explored (less than 1% literature contribution) with 5+ billion affected social media users, are. Moreover, single-axis adjustments for controlling bias in systems, for instance in hiring algorithms, fail to address the intersectional bias that, for example, black females suffer from in integrated gender/racial bias. There is thus essential differential research relevance for the 18 different types of bias that must consider that: Geographic, Disability, despite difficulty of application, demand immediate new-fabric research, Confirmation bias needs application-based Behavioral Research, while intersectional analysis must expand past race/gender bias, including analyses on Disability, Economic, and Religious factors.

1.1. Motivation & paper selection

The increasing deployment of Natural Language Processing (NLP) systems in critical fields such as healthcare, recruitment, law enforcement, and education has amplified concerns regarding algorithmic bias. Although model performance and scalability have improved substantially with the advent of large language models (LLMs), extensive research has shown that these systems often encapsulate and magnify societal biases based on gender, race, age, disability, religion, geography, and political orientation. Such biases can lead to unjust outcomes, reinforce systemic inequities, and undermine public trust in AI technologies.

To provide a comprehensive and up-to-date perspective, we systematically screened over 250 research papers published between 2014 and 2025 across leading academic sources, including Google Scholar, arXiv, IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, and DOAJ. Following a rigorous filtering process based on methodological rigor, empirical contributions, and interdisciplinary relevance, a final set of 121 papers was chosen for in-depth review. These works specifically focus on the detection, measurement, and mitigation of bias in NLP systems.

In recent years, there has been a notable increase in research activity in this field. While early contributions were limited (with only one paper in 2014 and another in 2016), steady growth began after 2019. The most notable surge occurred in 2024, with 87 papers published in a single year, marking the peak of scholarly interest in this field. Although the count for 2025 is based on partial data, it already indicates sustained momentum and continuing academic engagement. This pattern highlights both the timeliness and urgency of bias research in NLP, particularly in the age of large language models.

A comprehensive analysis of bias throughout the NLP pipeline, including data collection and annotation, model training, evaluation, and deployment, is made possible by this meticulously selected and varied

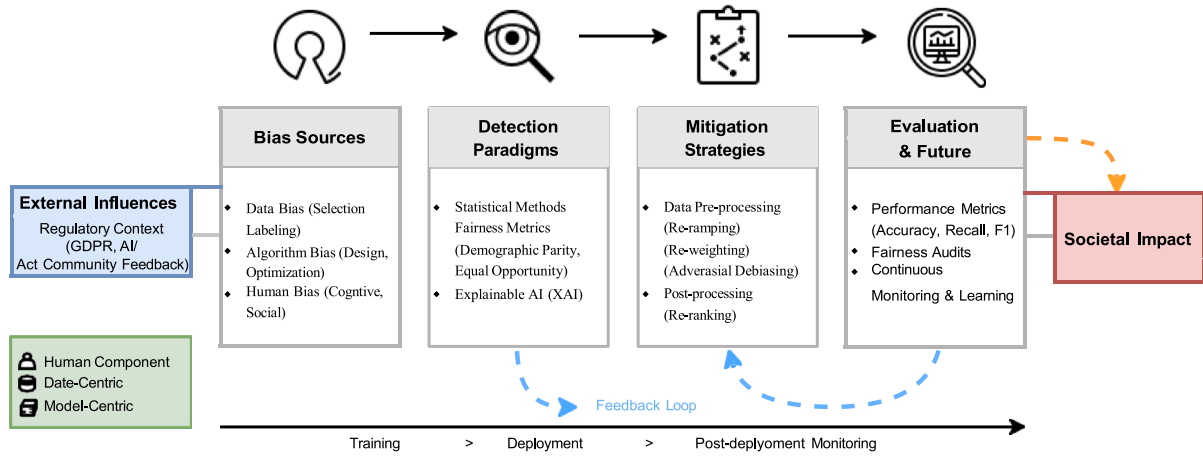


Fig. 1. Overview of the integrated NLP bias framework.

corpus. By synthesizing insights from technical, social, and applied perspectives, this review aims to advance the development of NLP systems that are more transparent, equitable, and socially responsible.

1.2. Contribution

By offering a distinctive and thorough synthesis of the field, this study significantly advances NLP fairness. Our work offers a structured framework illustrated in Fig. 1 with new perspectives, specifically contributing through:

- **A Novel and Structured Taxonomy:** We propose a systematic framework that categorizes both major and minor biases (e.g., data, linguistic, cultural) and their interconnections, providing a clear lens for researchers and practitioners.
- **Holistic Synthesis of Methods:** We provide a complete pipeline perspective by connecting state-of-the-art methods for bias detection and mitigation across data pre-processing, model-level adjustments, and post-processing.
- **Identification of Critical Research Gaps:** Our analysis identifies key challenges in underexplored areas, such as the scalability of debiasing for LLMs, intersectionality, and fairness-performance trade-offs, offering a roadmap for future research.

Distinction from prior bias taxonomies. Existing surveys on NLP fairness commonly organize bias using high-level categories such as label bias, selection bias, semantic bias, or model bias. While these abstractions provide valuable conceptual grounding, their coarse granularity limits their practical utility when diagnosing compound or deployment-stage harms. In contrast, our contribution lies not in redefining bias theory but in operationalizing these abstractions into a fine-grained taxonomy of 18 bias types that are explicitly mapped to stages of the NLP lifecycle. By disaggregating high-level bias sources into actionable subtypes, such as geographic, disability, annotation, evaluation, and deployment bias, our framework enables clearer diagnosis of where bias emerges and which mitigation strategies are most appropriate. This lifecycle-aware structuring supports interpretability and intervention planning beyond what prior hierarchical frameworks typically provide.

To contextualize the research landscape and identify strategic gaps, Table 2 maps each bias type across three dimensions: the population affected, how difficult it is to detect, and the current maturity of mitigation techniques. From this analysis, four distinct research priority levels emerge: Critical biases combine high impact, hard detection, and nascent mitigation; High priority biases show moderate-to-high impact with medium-hard detection and developing solutions; Balanced biases indicate that current research attention is proportional to their societal impact and mitigation maturity; and Medium priority biases reflect

moderate levels of impact, detection difficulty, and developing mitigation. The table shows that current research is dominated by gender and political bias, while other high-priority types like geographic, religious, and age biases remain underexplored.

Furthermore, Table 3 contrasts our study with recent reviews, highlighting how we advance the field with a more granular taxonomy of 18 bias types, a lifecycle-aware synthesis, and up-to-date coverage through 2025, establishing a structured, actionable, and forward-looking framework for NLP fairness.

2. Methodology: Systematic Literature Review (SLR)

This study employs a systematic literature review (SLR) guided by the PRISMA 2020 framework to ensure transparent, reproducible, and interdisciplinary synthesis as depicted in Fig. 2.

A key limitation arises from the predominance of English-language and Western-context studies, reflecting a global imbalance that exemplifies the “Generalization” challenge discussed in Table 4.

The review addresses four guiding research questions: (RQ1) the types of bias studied in NLP systems, (RQ2) methods for bias detection and measurement, (RQ3) effective mitigation strategies across the ML pipeline, and (RQ4) domain-specific challenges in areas such as healthcare and hiring. A comprehensive search (2014–2025) was conducted across major academic databases Scopus, Web of Science, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, DOAJ, ACL Anthology, and arXiv using Boolean combinations of core, action, and bias-specific keywords as, as detailed in Appendix A (Table A.12). Filters included publication date, English language, and empirical or theoretical contribution criteria. The multi-stage screening reduced over 250 initial results to a final corpus of 121 rigorously selected papers, ensuring methodological and thematic diversity.

2.1. Quality and bias assessment of reviewed studies

To strengthen methodological rigor, we conducted a qualitative quality assessment of the included studies following PRISMA 2020 recommendations. Each paper was evaluated along four dimensions: (i) clarity of bias definition, (ii) dataset transparency and representativeness, (iii) robustness of evaluation metrics, and (iv) reproducibility of methods.

Our analysis reveals several recurring limitations. A substantial portion of studies rely on single-metric evaluations (e.g., WEAT) without triangulation across tasks or datasets. Many evaluations are confined to English-language corpora, limiting generalizability to low-resource languages. Additionally, few studies assess bias persistence after deployment, highlighting a gap between offline evaluation and real-world performance.

Table 2
Research impact analysis of 18 bias types in NLP.

Bias type	Total papers (%)	Population affected	Detection difficulty	Mitigation maturity	Research priority	Ref.
Gender	11%	3.8B women	Easy	Mature	Balanced	[12,21–23]
Racial	5%	2B+ minorities	Easy	Developing	Balanced	[24–27]
Geographic	5%	4B+ Global South	Hard	Nascent	Critical	[18,28,29]
Disability	5%	1.6B disabled	Hard	Nascent	Critical	[30–34]
Political	8%	3B+ polarized regions	Medium	Developing	Balanced	[35,36]
Confirmation	<1%	5B+ users	Hard	Nascent	Critical	[37–39]
Religious	3%	2B+ minorities	Hard	Nascent	High	[40–42]
Linguistic	4%	3B+ non-English speakers	Medium	Developing	High	[43–45]
Cultural	4%	2B+ non-Western	Medium	Developing	High	[46–48]
Age	2%	1B+ older adults	Medium	Developing	Medium	[49–51]
Economic	2%	2B+ low-income	Medium	Developing	Medium	[52–54]
Adversarial	2%	All domains	Hard	Nascent	High	[55,56]
Evaluation	2%	Affects all fairness measurement	Hard	Nascent	High	[57,58]
Annotation	4%	Training data quality	Medium	Developing	Medium	[16,17]
Data Bias	4%	Training distributions	Easy	Mature	Balanced	[59–61]
Model/Architecture	3%	All models	Easy	Mature	Medium	[62–64]
Overfitting	3%	Generalization failures	Easy	Mature	Medium	[50,65–67]
Training (Selection)	3%	Optimization biases	Medium	Developing	Medium	[21,22]

Table 3
Comparison of key NLP bias review papers with our study.

Year	Contribution/Key findings	Research gap/Limitation
2020 [6]	Seminal survey highlighting that “bias” is often underspecified and techniques are poorly matched to motivations. Advocates for grounding work in social science literature on language & power.	Primarily conceptual critique; lacks a structured framework for evaluating metrics or practical utility.
2021 [11]	Introduces a formal, ontology-based definition of stereotyping bias in LMs, framing it as deviations in probability distributions for protected attributes. Reviews technical mitigation like vector space manipulation.	Rigorous but difficult to operationalize for real-world harms. Emphasizes technical fixes over socio-technical processes.
2022 [68]	Broad survey categorizing sources of bias (Label, Selection, Amplification, Semantic) and debiasing approaches (Pre-/In-/Post-processing). Summarizes common techniques such as Hard Debias and Gender Swapping.	Focuses heavily on gender bias and older, non-contextual metrics (e.g., WEAT). Limited critique of metric design vs. real-world impact.
2022 [69]	Applies NLP for automated risk-of-bias assessment in preclinical animal studies (randomization, blinding). CNN models outperform rule-based methods (e.g., 82% F1 for random allocation).	Focuses on methodological bias in biomedical research, not societal bias. Findings may not generalize to NLP fairness context.
2024 [4]	Introduces and formalizes the concept of <i>actionability</i> for bias measures. Defines desiderata (Motivation, Construct, Interval/Ideal, Use, Reliability) and analyzes 146 papers against them.	Limited to ACL Anthology papers. Actionability definition is context-dependent and may miss broader ML perspectives.
2026 (Our Study)	Proposes a novel, highly granular taxonomy categorizing 18 bias types (economic, geographic, confirmation, adversarial). Synthesizes detection/mitigation strategies across the NLP lifecycle. Reviews 121 papers (2014–2025) and provides a “cookbook” linking bias types to technical countermeasures.	Lacks a unifying theoretical framework like actionability. Focuses on breadth over depth. Mitigation strategies are listed but not critically evaluated for real-world trade-offs.

Table 4
Cross-cutting challenges in bias detection.

Challenge	Description	Bias types affected	Ref.
Intersectionality	Detecting biases that simultaneously affect individuals with multiple marginalized identities.	Racial, Gender, Disability, Age	[32,34,53]
Context dependence	Identifying bias expressions that are subtle and highly dependent on linguistic, cultural, or situational context.	Cultural, Linguistic, Disability, Political	[31,35,46,70]
Metric actionability	Interpreting statistical bias metrics that reveal disparities without providing actionable guidance for mitigation.	Nearly all types, but specifically discussed in evaluation frameworks.	[4,9]
Generalization	Applying bias detection methods across languages, cultures, or domains where performance often degrades.	Cultural, Linguistic, Geographic, Religious	[28,41,71,72]

Rather than excluding studies based on these limitations, this assessment provides context for interpreting existing findings and underscores the need for standardized, multi-dimensional evaluation protocols in future NLP fairness research.

Quantitative distribution of bias research. To systematically derive research gaps, we analyzed the final corpus of 121 studies according to their primary bias focus. As summarized in Fig. 3, research attention is unevenly distributed across bias types. Gender bias dominates the literature approximately 11% of studies, followed by political bias 8% and racial bias 5%. In contrast, disability, geographic, and annotation bias each account for roughly 5% or fewer of reviewed works, while age and religious bias are addressed in fewer than 3% of studies.

This imbalance quantitatively substantiates the research gaps identified in this review. Despite their significant real-world impact, biases affecting disabled populations, low-resource regions, and annotation processes remain substantially underexplored. These findings motivate a shift in future NLP fairness research toward broader demographic coverage and lifecycle-aware evaluation.

Temporal analysis showed that there was minimal work done before 2016, followed by rapid growth post-2019 and a publication peak in 2024, signaling increasing urgency in fairness research. The curated corpus supports lifecycle-based categorization of NLP bias covering dataset imbalance, annotation subjectivity, embedding and training bias, fairness assessment, and deployment monitoring capturing

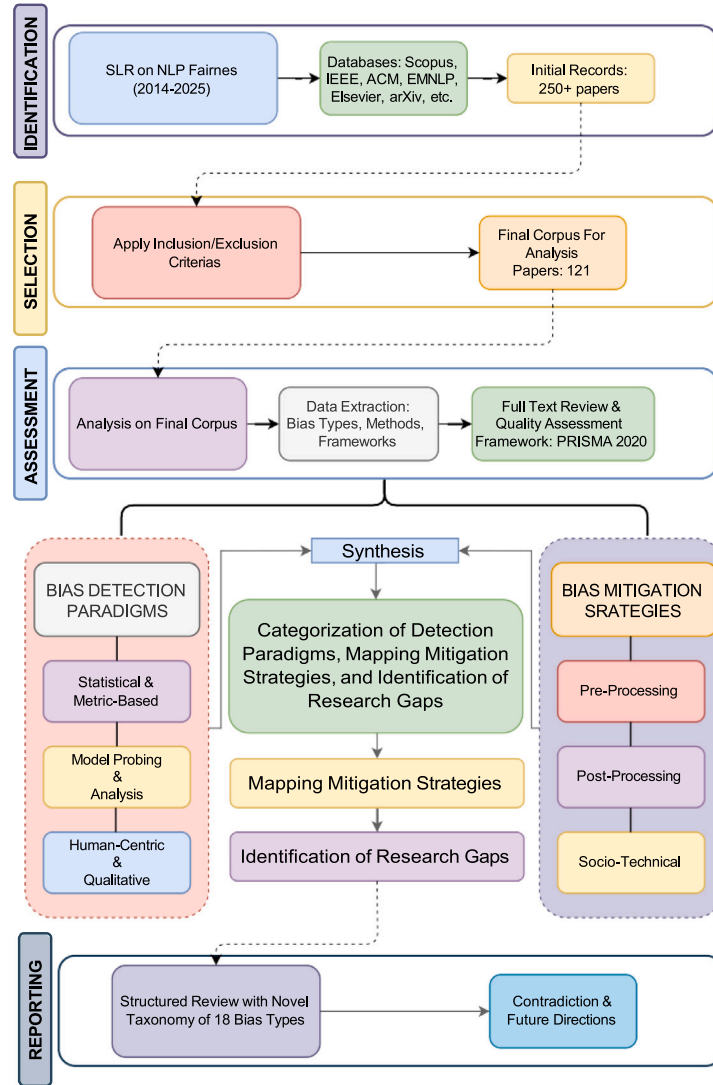


Fig. 2. Systematic review process followed for this study.

both technical mechanisms and socio-technical implications essential to responsible NLP research.

3. Bias types and detection methodologies

We structure our exploration of detection methodologies around a taxonomy of four major paradigms, as shown in Fig. 5: statistical & metric-based, model probing & analysis, benchmark & audit-based, and human-centric & qualitative approaches.

To structure our analysis, we adopt a four-paradigm taxonomy of bias detection approaches, as outlined in Table 5. This framework guides our discussion of techniques across statistical, model-probing, benchmark-based, and human-centric methods.

While our primary focus has been on identifying and analyzing biases in NLP systems, developing effective mitigation strategies is equally crucial for creating fair and equitable models. Fig. 4 illustrates the lifecycle of bias in NLP systems, showing key stages where bias can emerge and be mitigated. Table 6 provides a concise summary of recommended mitigation techniques for the major bias types discussed in this review, which we will now explore in detail.

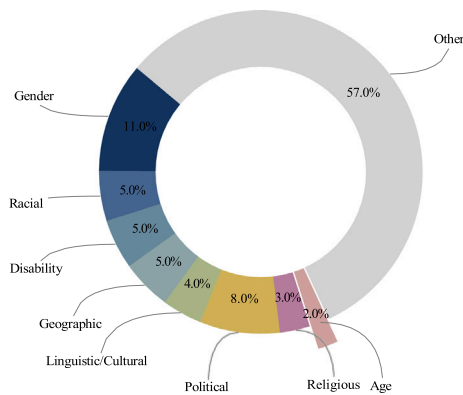


Fig. 3. Approximate distribution of top bias types in reviewed papers.

Table 5
High-level taxonomy of bias detection paradigms.

Detection paradigm	Description	Example techniques	Applicable bias types (Examples)	Ref.
Statistical & Metric-based Model probing & Analysis	Uses quantitative metrics to measure disparities in model outputs, performance, or representations. Examines the internal mechanisms of models (e.g., embeddings, attention) to uncover latent biases.	p-% Rule, Statistical Parity, WEAT/SEAT, Perplexity Comparison, Perturbation Sensitivity, Masked Language Modeling, Latent Embedding Analysis	Racial, Gender, Age, Model, Economic Disability, Linguistic, Adversarial, Religious	[10–12,24,62] [30,55,73,74]
Benchmark & Audit-based	Uses carefully designed datasets or templates to systematically test for specific biased behaviors.	Template-based tests (BITS), Challenge datasets (CrowS-Pairs), Algorithmic Audits	Disability, Geographic, Gender, Political	[25,35,73,75]
Human-Centric & Qualitative	Incorporates human judgment, expert review, or community feedback to identify nuanced or contextual bias.	Thematic Coding, Expert Review, Participatory Design, Matched Guise Tests	Cultural, Disability, Annotation, Deployment	[25,31,76,77]

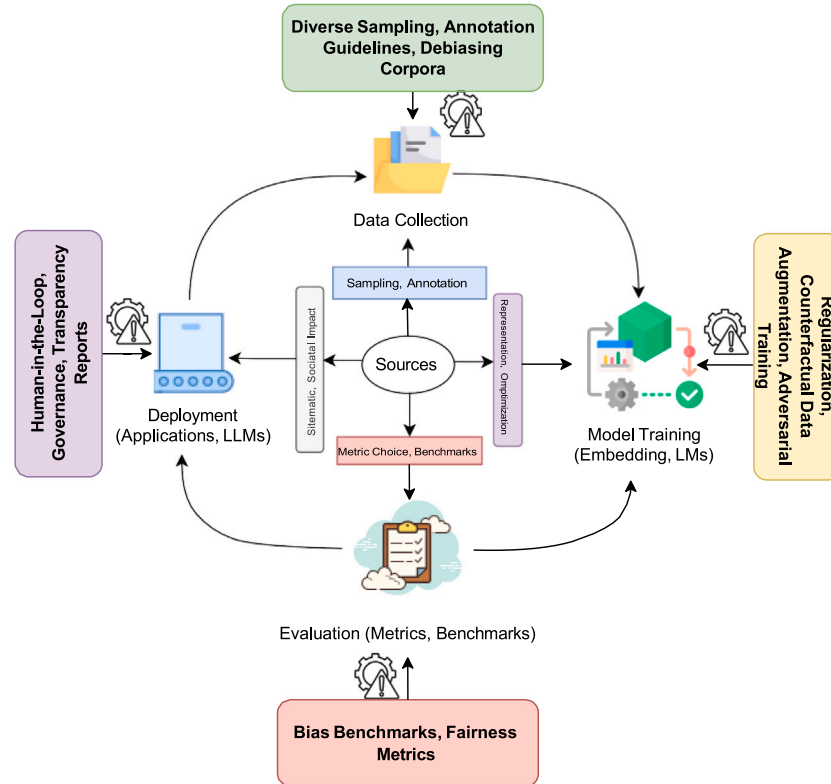


Fig. 4. Lifecycle of bias in NLP systems, showing key stages where bias emerges and can be mitigated.

3.1. Case study: Lifecycle diagnosis using the 18-type taxonomy

To demonstrate the practical utility of the proposed taxonomy, we consider a representative NLP application: toxicity detection using large language models deployed on social media content.

Using conventional bias frameworks, observed disparities in toxicity scores for content authored by speakers from the Global South might be attributed broadly to data or model bias. However, applying our taxonomy reveals multiple interacting bias types across the NLP lifecycle.

At the data level, geographic bias emerges due to the underrepresentation of non-Western dialects and regional language varieties in training corpora. During annotation, annotation bias arises from culturally misaligned labeling guidelines and annotator pools lacking regional diversity. At the evaluation stage, evaluation bias manifests when benchmark datasets fail to reflect linguistic variation, leading to misleading fairness assessments. Finally, during deployment, deployment bias becomes evident as real-world user data diverges from benchmark distributions.

By explicitly distinguishing these bias types, the taxonomy clarifies why post-hoc mitigation alone (e.g., output filtering) is insufficient. Instead, it motivates multi-stage interventions, such as region-aware data augmentation, culturally grounded annotation protocols, and continuous post-deployment audits. This example illustrates how the proposed taxonomy enhances interpretability and supports more precise bias mitigation strategies compared to prior coarse-grained frameworks.

3.2. Sociodemographic biases

3.2.1. Racial bias

Racial bias in NLP systems manifests as disparate treatment and stereotyping of racial groups, requiring comprehensive detection and mitigation approaches. Detection techniques consist of dataset analysis for implicit biases related to underrepresented groups, homogeneity metrics that identify stereotyping representations, and repositories of stereotypes to track systematically [25].

Statistical methods rely on Word Embedding Association Test (WEAT), a method measuring implicit biases in embeddings by assessing association strength between target and attribute words [12].

Table 6
Mitigation strategies and implementation considerations for major types of bias in NLP.

Bias type	Mitigation techniques	Implementation considerations & Potential pitfalls	Ref.
Racial bias	Diversified dataset curation; Counterfactual data augmentation; Community involvement in development; Fairness constraints in optimization.	When curating diverse data, ensure representational balance <i>without tokenism</i> mere inclusion of minority data does not guarantee fairness if labeling remains biased. Counterfactual augmentation may create unnatural sentences; verify linguistic plausibility via human review. Community involvement should include marginalized stakeholders early in data design, not only post hoc.	[25]
Gender bias	Counterfactual data substitution; Adversarial debiasing; Rescaling word embeddings; Gender-synchronized training data; Post-hoc algorithms (e.g., 2DAMA).	Adversarial training can reduce performance if the adversary is overfitted tune the loss balance (λ) carefully. Counterfactual substitution can leak artifacts if gender markers (e.g., “she” → “he”) do not preserve context. Evaluate debiasing effectiveness using both intrinsic (e.g., WEAT) and extrinsic (task) metrics.	[21–23]
Disability bias	Participatory design with disabled communities; Contextual model design; Counterfactual data augmentation; Inclusive dataset curation (e.g., BITS corpus).	Requires long-term collaboration with disability advocates; token consultation is insufficient. Beware of over-representation of “medical” narratives balance social and functional perspectives. Ensure accessibility of annotation tools for disabled contributors.	[31,32,73]
Political bias	Active monitoring and auditing of LLM outputs; Transparency in training data sourcing; Ethical design with human-in-the-loop oversight; Collaborative open-source efforts.	Political bias mitigation demands <i>temporal monitoring</i> models drift as political discourse evolves. Ensure annotators from multiple ideological backgrounds. Avoid post-hoc censorship which may introduce new bias.	[36]
Geographic bias	Fine-tuning on diverse, multilingual geographic data; Stratified topic modeling; Use of spatially balanced corpora (e.g., GeoWiki); Parameter-efficient fine-tuning for few-shot geocoding.	Cross-lingual fine-tuning may still favor dominant languages if tokenizers are skewed. Verify geographic coverage statistically (e.g., HDI distribution). Use region-aware evaluation metrics such as MdnED.	[28,29,75]
Annotation bias	Double annotation with conflict resolution; Clear and detailed annotation guidelines; Diversification of annotator pools; Combining data from multiple sources.	Pay annotators fairly; bias may stem from socioeconomic status. High inter-annotator disagreement can indicate under-specified labels revise guidelines iteratively. Record demographic metadata for transparency.	[16,17]
Model bias	Adversarial debiasing; Re-sampling techniques (e.g., SMOTE); Post-processing (e.g., threshold adjustment); Use of bias detection toolkits (e.g., AIF360).	Toolkits require correct protected attribute labeling errors propagate. Post-processing may improve fairness but hurt calibration; validate on holdout sets. Document trade-offs transparently.	[62,63]
Data bias	Data augmentation (back-translation, synonym substitution); Oversampling minority classes (SMOTE); Data diversification; Fairness constraints.	Back-translation can distort meaning in low-resource languages; human review is advised. Oversampling can cause overfitting on minority samples prefer dynamic reweighting.	[60,61,78]

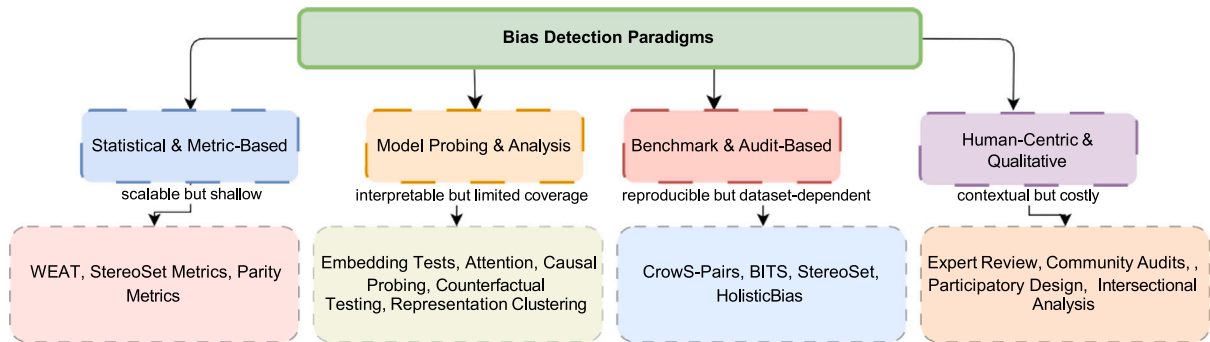


Fig. 5. Taxonomy of bias detection paradigms in NLP, illustrating four major categories and representative techniques.

The p -% Rule expresses a word’s familiarity as a degree between 0 and finds disparities in outcomes, such as reduced positive prediction rates for blacks (e.g., 28%) [12]. Other techniques involve log-odds analysis, coreference resolution with comparison results [24], and specialized methods for African American Vernacular English (AAVE) bias detection [26,27].

Mitigation strategies focus on dataset diversification with culturally representative data, counterfactual augmentation to balance group representation, and data filtering with reweighting during fine-tuning. Community involvement throughout development enhances cultural sensitivity, while multiracial testing environments and transparent algorithmic practices ensure accountability [25].

3.2.2. Age bias

Age bias detection employs multiple methodological approaches. Pretrained models like XLM-RoBERTa fine-tuned on annotated comments effectively identify age-based hate speech and slurs across languages [49], with ensemble methods like ModelWardrobe enhancing

robustness through multiple model variants [49]. Latent Dirichlet Allocation (LDA) extracts subtle age-bias indicators from annotated data, while manual annotation provides real-world grounding for detection capabilities [49].

Statistical methods include perplexity analysis to reveal disparities in language model probabilities for age-specific terms [50], complemented by ANOVA and t-tests to quantify behavioral differences across demographic groups [50]. Manual annotations validate findings and inform bias specification dictionaries targeting age-related lexical triggers [50].

Embedding association tests quantify bias through WEAT for word-level associations [10] and SEAT for sentence-level representations [10], while Bias-NLI frameworks detect inferential biases using neutral premise-hypothesis templates [10]. Comprehensive LLM surveys combine qualitative case studies with quantitative fairness metrics like statistical parity and equalized odds [51].

Tools like BiasAlert enable continuous human-in-the-loop monitoring [51], while demographic distribution analysis and domain adaptation techniques identify performance disparities and feature representation inconsistencies across age groups [79]. Domain adaptation helps detect whether models maintain invariant feature representations across age demographics [79].

3.2.3. Religious bias

Religious bias detection employs diverse methodologies across languages and contexts. Kadan et al. utilize corpus-level affective bias analysis examining affective word distribution imbalances, alongside prediction-level measurements with controlled sentence pairs to reveal systematic emotional prediction disparities, using public evaluation corpora like EEC, BITS, and CSP for quantitative assessments [40].

Farrelly et al. leverage transformer embeddings from BERT and RoBERTa with K-Nearest Neighbors classification and t-SNE visualization for bias cluster identification [71]. Sadhu et al. apply template-based and naturally sourced probing for Bangla language models, quantifying bias with Disparate Impact scores [41]. Fauzan et al. use generalized Pointwise Mutual Information (PMI) to identify associations between religious terms and sentiment labels in Indonesian datasets [80].

KhudaBukhsh develops toxicity auditing frameworks including the toxicity rabbit hole method to reveal hidden biases in LLMs, employing polyglot language identification for multilingual social media texts [42]. Donald et al. survey customer interaction data using sentiment analysis, emotion recognition, and association tests like WEAT for demographic bias quantification [81].

Mitigation approaches include bias-aware dataset curation with balanced affective word distributions [40], transformer embeddings for targeted interventions [71], retraining on culturally nuanced datasets with safety guardrails [41], neutral dataset augmentation to balance sentiment labels [80], counterbalancing frameworks with toxicity-based moderation [42], and transparent documentation through model cards combined with fairness constraints [81].

3.2.4. Disability bias

Disability bias detection employs diverse methodologies. Hassan et al. use sentiment analysis and Hierarchical Dirichlet Process (HDP) topic modeling to identify negative emotional valence and themes like “death” and “pain” in disability-related outputs, validated statistically with ANOVA and t-tests [32]. Venkit et al. develop perturbation sensitivity analysis and the BITS benchmark for template-based testing, showing models rate disability-related sentences as significantly more toxic [73].

Phutane et al. apply mixed-methods using tools like Google Perspective API and GPT-4 alongside human evaluations from disabled individuals, revealing LLMs lack nuanced understanding of disability experiences [33]. Gadiraju et al. conduct qualitative research identifying subtle harms like infantilization and inspiration porn through focus groups with disabled participants [31].

Gupta et al. survey comprehensive detection techniques including WEAT, ECT metrics, and prompt-based evaluations for stereotype exposure [34]. Venkit et al. further probe BERT’s internal biases through masked language modeling, finding negative completions for disability-related prompts [30]. Dobesh et al. demonstrate educational approaches using structured probing methods to surface disability bias in classroom settings [82].

Mitigation strategies include intersectional analysis frameworks and customized embedding debiasing [32], contextual model design and inclusive dataset curation with benchmarks like BITS [73], fine-tuning with disability-authored explanations and participatory design [33], community-defined annotation guidelines and open-source development [31], counterfactual augmentation and adversarial training techniques [34], pre-deployment auditing and inclusive corpora creation [30], and bias-reduced fine-tuning with educational interventions [82].

3.3. Linguistic and cultural biases

3.3.1. Geographic bias

Belliardo et al. (2023) developed a dual NER–geocoding pipeline combining SpaCy, roBERTa, and a custom FeatureRank algorithm to resolve toponyms from country- to facility-level mentions. Their analysis revealed overrepresentation of Western regions in HDI-based evaluations, highlighting systemic bias [28]. Mallick et al. (2024) used BERT-based NER and LocationTagger to extract and disambiguate toponyms in climate texts, aided by BERTopic and ChatGPT summarization for regional theme discovery. Their results exposed geographic underrepresentation in certain areas [29]. Mai et al. (2024) introduced a spatial representation learning framework integrating multi-scale, fairness-aware, and topology-preserving modules to mitigate regional disparities in spatial learning [83]. Bartl et al. (2025) surveyed gender and regional biases across NLP and CV, finding stronger linguistic bias in text models and higher correction costs in vision datasets [75]. Liu et al. (2022) quantified geoparsing bias using spatial metrics like Getis–Ord G_i^* , Median Error Distance, and Spatial Misalignment, exposing low recall in Global South regions [18]. Hu et al. (2024) fine-tuned lightweight LLMs (Mistral, Llama2) for toponym resolution via few-shot prompting and multi-geocoder querying, identifying region-dependent accuracy gaps [72].

Fine-tuning domain-specific models (e.g., SpaCy, roBERTa) improves geolocation precision by reducing false positives and enhancing disambiguation [28]. Incorporating multilingual and climate datasets further mitigates Western bias [29]. Spatially adaptive embedding learning reduces overfitting to regional mosaics [83]. Multimodal adversarial training helps curb stereotype propagation [75]. Spatially balanced corpora such as GeoWiki enhance global representation [18], while parameter-efficient fine-tuning and asynchronous querying sustain LLM accuracy across regions [72].

3.3.2. Cultural bias

Detecting cultural bias in NLP and LLMs involves analyzing language data, model behavior, and outputs through cultural perspectives. Linguistic and stylistic analyses reveal how dialects, politeness norms, or sociolects are undervalued in models trained on dominant varieties [46]. Evaluating shared cultural knowledge such as kinship terms or idiomatic expressions helps identify misinterpretations or exclusion of culturally specific content [46,76].

Quantitative detection uses perplexity comparisons and statistical tests (e.g., t-test, Cohen’s d) to measure performance disparities across cultural groups [47]. Lexical variance analysis with WED or TF-IDF highlights subtle vocabulary differences reflecting unequal cultural representation [76]. Model-centered evaluations employ fairness metrics like Equal Opportunity, interpretability tools (SHAP, LIME), and counterfactual fairness tests to uncover biased decision patterns [48]. Algorithmic audits testing NLP systems with culturally diverse or identity-specific inputs further expose biases often hidden from quantitative metrics [70].

Mitigation emphasizes culturally diverse data collection and inclusive annotation to stabilize model performance across demographics [46]. Balanced datasets and adaptive, culturally aware translation improve cross-cultural generalization [47]. Decolonized, participatory NLP design grounds models in local sociohistorical realities [76]. Counterfactual augmentation, fairness constraints, and post-model self-debiasing further reduce latent bias [48], while community-driven audits and iterative expert review foster fairness, transparency, and cultural accountability [70].

3.3.3. Linguistic bias

Linguistic bias arises when NLP models favor dominant languages or dialects, resulting in weaker performance on marginalized varieties. Equal Error Rate (EER) comparisons across corpora like VoxCeleb-1 and Aishell-1 reveal poorer generalization in non-English contexts, emphasizing training data imbalance [43]. Lexical diversity and bundle analyses of ACL and EMNLP abstracts expose regional stylistic disparities, showing bias in academic writing conventions [44]. Similarly, lexical database studies highlight inadequate representation of non-English translations, revealing systematic multilingual exclusion [84].

Feature annotation on minoritized dialects (e.g., Kenyan English) demonstrates reduced morphosyntactic retention in model outputs, quantifying inclusivity gaps [45]. Sentiment analysis of clinical notes uncovers biased adjective patterns and mistrust language toward minority patients [77,85]. Chi et al. (2025) show that high morphological complexity in Chinese text degrades model accuracy, which their hybrid BERT-TCN-BiLSTM-Attention model mitigates [86]. In hiring systems, confusion matrix analysis reveals gendered linguistic patterns despite balanced datasets [87].

Mitigation strategies involve model design and community participation. Frame-attentive architectures enhance cross-lingual embedding transfer with minimal fine-tuning [43]. Expanding underrepresented lexicons improves multilingual inclusivity [84], while stylistic guidance fosters fairness in academic writing [44]. Compassionate and bias-aware documentation supports equitable clinical NLP [77].

3.4. Data and model biases

3.4.1. Overfitting bias

Overfitting bias occurs when high-capacity models memorize training noise or quirks instead of learning general patterns, causing poor performance on new data and a big gap between training and test error. [88]. Zhang et al. identified instability in BERT fine-tuning under few-sample scenarios, attributing it to the use of the BERTAdam optimizer, which omits gradient bias correction. Detection methods include measuring the variance in test accuracy and analyzing layer specialization using L2 distance from initialization. Their experiments show that upper transformer layers tend to overspecialize, and performance varies heavily across random seeds [67]. Gururangan et al. focused on detecting domain mismatch in pretrained language models through domain similarity analysis using unigram overlap and masked language modeling loss. Poor vocabulary overlap (e.g., only 20% for BioMed vs. RoBERTa) indicates pretraining inadequacy. They also quantify cross-task performance decay during task-specific adaptation, flagging overfitting or misalignment [66]. Zhang et al. explored the ability of deep learning to memorize noise via several generalization stress tests, such as label randomization, input corruption (e.g., Gaussian noise, pixel shuffling), and partial label noise injection. These detection setups showed that even meaningless or corrupted data could be fit perfectly, revealing flaws in our understanding of generalization. Further, SGD's implicit bias was examined by analyzing convergence to minimum-norm solutions in linear models [65]. Tang et al. introduced a dendritic neural architecture designed for high-dimensional classification, evaluated on datasets like Breast Cancer and Parkinson. Although not explicitly biased, their use of cross-entropy loss, multi-output neurons, and validation across multiple datasets serves as implicit detectors of classification errors and structural model flaws [89]. Srivastava et al. proposed dropout as a regularization strategy and identified overfitting by monitoring validation loss trends and employing Monte Carlo model averaging over dropout-induced subnetworks. They detected overfitting when networks achieved low training loss but high test error. Experiments on MNIST and CIFAR-10 confirmed that dropout mitigates these discrepancies [90].

The problem of BERT fine-tuning instability has begun to be addressed with bias correction in the Adam optimizer and top transformer

layer re-initialization, along with longer training schedules and regularization methods, leading to a return toward more stable results [67]. The method described leverages continuous-domain nearest-neighbor data augmentation in combination with Domain-Adaptive Pretraining(DAPT) and Task-Adaptive Pretraining(TAPT), which first pretrain on unlabeled target domain data to obtain a more general representation that better matches the source domain [66]. Optimizer that is SGD in this case provides automatic internal regularization behavior which pulls solutions toward the lower-normed manifold of models; architecture decisions and domain-specific augmentations were found to provide more compelling improvements over explicit regularizers for generalization [65]. Custom dropout has also been used for enhanced dendritic neural networks to prune inactive synapses dynamically, where individual learning rates aid in better adaptation and overfitting reduction in high-dimensional settings [89]. Stochastic regularization like Dropout was introduced to various extend the work to different datasets and network types, leading to improvements in generalization as well as providing insights for future breakthroughs in training efficiency and theory [90]. Overall, overfitting bias undermines NLP reliability by causing models to memorize noise rather than learn general patterns, but techniques like dropout and stress testing effectively detect and mitigate it through validation monitoring and regularization

3.4.2. Model bias

Model bias is indicated by algorithmic differences in outcomes. Statistical parity in statistical analysis gives equal predictions, and chi-squared tests identify dataset biases [62]. Diagnostic performance is assessed by clinical predictive value analysis, and epidemiological studies uncover biases [62]. Data collection processes are compared by prospective studies and biases are discovered. Policies are analyzed by auditing tools like the Disparities Impact Statement, while the PROGRESS framework quantifies inequality factors [62]. Implicit Association Test, dashboards, Equity Quality Improvement Tool, and Cultural Competence Assessment offer equitable frameworks [62]. Feedback from end-users identifies bias in clinical decision-making [62]. Confusion matrices in employee selection algorithms review performance across protected groups, with step-by-step analysis identifying high-bias words [63].

Model bias that occurs from training or design can be minimized via balanced data, fair algorithms, and non-technical interventions. These methods include re-sampling (such as SMOTE) methods, which amplify the salience of underrepresented groups in ML [62], and adversarial debiasing aims to train predictors and adversaries together to reduce the influence of protected features in ML [63]. Post-processing fixes, such as threshold re-tuning, are tuned to align outputs with demographic characteristics used in applications like sepsis prediction [62]. Non-technical solutions, including blind recruiting based on resume, diversified screening panels, reduce the recruitment bias [63]. Such frameworks, such as the AI Fairness 360 of IBM, allow detecting bias in a platform used to make decisions and addressing it [63].

3.4.3. Annotation bias

Human annotations are biased; annotations may not reflect the truth about the dataset. IAA measures the consistency of the annotator by using F measures, and shows bias if the annotators are not reliable with their preannotations [16]. This method allows us to measure agreement to spot any subjective effects. Scores that are not high enough for IAA often show that the dataset was created with unclear annotation practices. Due to its utilization in medical and social media datasets, the approach ensures strong evaluation. It is used in quality control, which allows for better annotation modifications. The results suggest that there is a drop in performance during the 'preannotation' period that can later be corrected and improve the performance of the annotators [16]. ANOVA looks at both IAA and performance differences, while examining each file to find practice inconsistencies [16]. Causal Mediation Analysis measures how neurons shape biases and

finds what specifically mediates the bias [74]. Embeddings obtained from counterfactual generation are biased, and analysis of behavior reveals a dependence on surface patterns [74]. SPARQL queries on knowledge graphs identify annotation patterns, disentangling visual transformer biases [91]. The community detection process matches annotators by their background and analyzes which classifiers achieve the best results [91]. To find out how consistent the annotation is, experts use measures called confusion matrices and compare the resulting data with LLMs [17]. Real-world annotation analysis illustrates political leaning affecting accuracy, with fairness metrics measuring error disparities [59]. Facial Action Units detect gender-based biases in expression annotations, with chi-square tests and logistic regression confirming differences [64].

Annotation bias through imbalance or subjective dataset labeling judgment can be reduced through systematic annotation practices and varied data management. Double annotation, two annotators labeling data independently and conflicts being settled under supervision, increases consistency [16]. The combination of data across sources like Reddit and other social media introduces models to multiple linguistic styles, decreasing the chance of bias since annotations look similar [17]. Clear annotation guidelines, which discuss annotation parameters such as direct mention or aggressive tone, are helpful in encouraging consistency and minimizing subjectivity [17]. The use of annotators with different backgrounds will be opened, and having more than one annotation per item increases reliability by interjecting different judgments [17].

3.4.4. Data bias

Data bias, an imbalance in data distribution, harms model performance and fairness. Minority class percentages reveal mild (20%–40%), moderate (1%–20%), or extreme (<1%) imbalances, e.g., 11.9% for “sense of responsibility” flags bias [78]. Class distribution analysis identifies disparities affecting underrepresented groups in tasks like sentiment analysis [78].

In material science, density distribution (atomic weight/crystal volume) exposes bias, with non-synthesizable samples showing greater density ranges [60]. Mean atomic mass and volume distributions reveal clustering biases, aiding imbalance detection [60]. Cross-evaluation on swapped test sets detects bias via performance drops (e.g., 0.967 to 0.51), indicating spurious correlations [60]. Compression tests (e.g., 55% volume reduction) uncover density-driven biases in mixed-source models [60].

Manual annotation (e.g., 200 MAB-English samples) with Jaccard similarity and credibility scores detects subtle biases (e.g., race/gender) better than automated methods [92]. AI Fairness 360 metrics (e.g., German Credit Dataset) quantify fairness via consistency (0.681), disparate impact (0.8338), and mean difference (−0.1185) [13].

Intrinsic bias testing uses masked language evaluation to identify latent biases in pre-trained models [61]. Extrinsic metrics (e.g., word embedding tests, sentiment analysis) reveal group performance disparities [61]. These methods offer a comprehensive bias detection framework [61].

Data bias issues can be resolved through data diversification and training process enhancement. A pre-processing technique can be dropping sensitive information or data augmentation to represent better, as in the case of music and film recommendations [61,93]. Data sets can lead to better diversity through back-translation and synonym substitution, as can be done on, for example, IMDB [61]. Merging data sets of diverse sources equalizes features in domains with specialization, such as predicting crystal compounds [60]. Oversampling of minor classes through techniques such as SMOTE of internal training data is a way of restoring the balance without compromising the test integrity [78]. Fairness constraints on training regulate the behavior of models on sensitive characteristics, whereas post-processing procedures such as re-ranking foster fairness at the end [61,93].

3.4.5. Training bias

Identity Term Sampling (ITS) detects gender bias in NLP training data by analyzing gendered term frequencies (e.g., woman/man, she/he). Categorizing samples as No-Gender (NG), Equal-Gender (EG), Positive Gender (PG), or Strongly Positive Gender (SPG), ITS achieved 91.8% validation on BiasBios, revealing high bias in Hate Speech datasets but minimal bias in Hotel Reviews [94].

The Identity Term Template Approach (GBETs) generates synthetic test data (e.g., “You are a <adjective><identity term>”) to measure gender bias. Equal male/female samples expose performance disparities, particularly in Hate Speech datasets, offering a rigorous bias evaluation tool [94].

In computer vision, PlantVillage 8px Training Strategy detects capture bias by classifying plant disease images using only 8 background pixels, achieving above-random accuracy, highlighting bias from irregular data collection [95]. In contrast, the MNIST 8px tests show a near-random performance, confirming background variability as a bias source [95]. Blur-based analysis further proves that bias persists even when isolating foreground objects [95].

For dataset refinement, Irrelevant Detection uses InceptionV3, MobileNet, and ResNet-50 to filter out-of-scope images via majority voting, reducing noise [96]. Duplicate Detection employs locality-sensitive hashing (LSH) to identify and remove duplicates, preventing skewed model performance [96].

These methods collectively address training bias across the NLP and CV domains.

3.5. Contextual and systemic biases

Biases in NLP systems are not limited to data, models, or annotation processes; they also emerge from broader contextual and systemic factors. These include biases introduced by political orientation, evaluation methodologies, regulatory frameworks, and societal contexts in which NLP models are deployed. Unlike technical biases, which can often be traced to specific datasets or algorithms, contextual and systemic biases are more deeply embedded in social structures and evaluation practices, making them harder to detect and mitigate.

3.5.1. Political bias

Political bias in NLP systems reflects ideological skew in text outputs. Classical approaches use Bag-of-Words and TF-IDF features with multinomial logistic regression to classify texts by ideology, enabling quantitative assessment of partisan tendencies in speeches or manifestos [97]. Deep learning methods like CNNs and bidirectional LSTMs capture contextual dependencies in political text, identifying ideological markers across news sources with stable cross-validation [98].

SentenceBERT-based stance detection measures cosine similarity between model outputs and ideological anchors, while framing and style bias analyses apply zero-shot classification and polarity estimation to detect subtle content skew [35]. Survey-driven studies such as Wahl-O-Mat and ANES-based frameworks evaluate LLM responses across political questions, quantifying ideological drift and language disparities [36,99]. The Political Compass Test complements these by probing response consistency under paraphrasing and negation, revealing latent political leanings [100].

RNN-based sentiment analysis further detects bias in model-generated essays and policy-related datasets, linking prompt design and temperature variation to ideological alignment [101]. Mitigation strategies emphasize transparency, open-source collaboration, and ethical oversight, supported by active monitoring and human-in-the-loop governance to ensure democratic accountability in NLP model design [36,98].

3.5.2. Evaluation bias

Evaluation bias arises when the assessment of a model's performance is distorted due to unrepresentative datasets, unsuitable metrics, or biased evaluation procedures. Such bias can mislead developers by overstating a model's capabilities or fairness, resulting in systems that fail to perform equitably across diverse populations. It underscores the importance of designing evaluation frameworks that accurately reflect real-world use conditions and population diversity.

Recent advances highlight structured attempts to diagnose and quantify evaluator bias. The LLMBench benchmark systematically uncovers position, verbosity, formality, and self-enhancement biases by employing adversarial datasets that reveal subtle distortions in model evaluation outcomes [57]. Similarly, COBBLER introduces a meta-evaluation framework that measures six cognitive biases both implicit and explicit through round-robin pairwise comparisons, leveraging statistical significance tests and Rank-Biased Overlap (RBO) scores to capture inter-evaluator agreement [58].

Further analyses, such as those based on SummEval, identify familiarity, anchoring, and round-number biases, as well as self-inconsistency measured using Krippendorff's alpha [102]. These findings reveal that evaluator judgments can be skewed by previous ratings, linguistic familiarity, or rounding tendencies, complicating the interpretation of model performance.

Cross-model comparisons further demonstrate architecture-specific disparities. Etaawi and Alhijawi [103] report that DeepSeek surpasses ChatGPT by 6.73% in classification accuracy across identical benchmarks, while ChatGPT marginally excels in generative fluency indicating that fairness evaluations are model-dependent and not fully generalizable. Beyond automated systems, bias detection has also been applied to human-generated content; for instance, analyses of over 70,000 faculty evaluations using Random Forests, GBM, and logistic regression uncovered gender and intersectional biases in academic assessments [104].

Mitigation strategies emphasize improved prompt engineering and evaluator configuration. Stureborg et al. [58] propose structured "evaluation recipes" that define metrics explicitly, encourage critical reasoning, and align model outputs with expert human ratings. These approaches, along with multi-model benchmarking and human-in-the-loop oversight, enhance reliability and transparency in LLM-based evaluation pipelines [58,102].

3.5.3. Deployment bias

Deployment bias arises when models performing well in development fail in real-world settings due to differing environments, data, or user behavior, reducing fairness and trust. Detection and monitoring employ statistical metrics (e.g., equalized odds, Disparate Impact Ratio), rapid prototyping, toolkits like IBM AI Fairness 360 and Google's What-If Tool, audits, and documentation such as model cards [105–107]. Mitigation spans the ML lifecycle: pre-processing (data balancing, counterfactual augmentation) [108,109], in-training (fairness-aware loss, adversarial debiasing) [108], intra-processing (layer calibration, FAST) [110], and post-processing (calibrated thresholds, constrained decoding) [108], with concrete strategies for error-gap reduction, modular debiasing, and joint mitigation of social biases [109,111]. Toolkits like Fairlearn and IBM AIF360, along with frameworks such as FAIR-Frame, support standardized metrics and alignment of fairness with outcomes, while real-world deployment requires balancing fairness, accuracy, cost, and continuous monitoring [112–115].

3.5.4. Adversarial bias

Adversarial bias in AI systems arises when models exhibit unfair or distorted behaviors due to hidden correlations or manipulative perturbations. Detection requires diverse approaches across domains. In education, fairness metrics such as Demographic Parity, Equalized Odds, and Equal Opportunity identify disparities in predictions based on income or disability status, flagging potential adversarial

effects [15]. In recommender systems, Latent Embedding Analysis and Attacker Networks reveal whether protected attributes like gender are implicitly encoded, exposing representational bias even without explicit demographic inputs [55].

In vision models, the Reasonability Matrix visually identifies bias when spurious contextual cues (e.g., kitchenware or sports gear) disproportionately affect predictions of gender or occupation [116]. Gradient-based detection and input reconstruction techniques detect model sensitivity to adversarial perturbations near decision boundaries, highlighting robustness vulnerabilities in image classification tasks [56]. For NLP systems, Word and Sentence Embedding Association Tests (WEAT, SEAT, CAT) measure implicit stereotype encoding in language representations [11]. In medical AI, feature inconsistency and adversarial preprocessing uncover anomalous distributions in diagnostic imaging, signaling adversarial interference [117].

Mitigation methods combine fairness, adversarial learning, and feature-level corrections. Group-unaware modeling removes protected attributes but fails when proxies persist, whereas adversarial debiasing employs gradient reversal to eliminate demographic signals from embeddings [15,55]. Latent space bias mitigation uses denoising autoencoders to preserve task-relevant features while suppressing bias. GRADIA refines attention maps based on human feedback, steering vision models toward semantically relevant cues [116]. Defensive distillation, randomized smoothing, and adversarial training enhance robustness in high-risk applications [56]. In NLP, counterfactual data replacement, bias-aware loss functions, and word embedding debiasing reduce linguistic stereotypes [11]. Guided denoisers and improved pooling mechanisms further strengthen fairness and generalization in medical imaging [117].

3.5.5. Economic bias

In Vision-Language Models (VLMs), economic bias detection often centers on identifying in-context learning (ICL) deficiencies linked to data distributional features such as bursty distributions, skewed marginals, and dynamic meaning. Bursty clusters group semantically related clips, while skewed marginals emphasize rare verb-noun pairs, revealing model sensitivity to long-tail contexts. Dynamic meaning variations test model generalization across phrasal diversity, with ablation studies confirming each feature's contribution to eliciting ICL [52].

In Latin American NLP, bias analysis employs culturally grounded lexicons (e.g., *madre* vs. *padre*) and concordance frequency studies, complemented by visual tools such as EDIA for distributional inspection. Sentence-level probes distinguish stereotype-reinforcing from stereotype-defying patterns, while intersectional analyses expose compounding effects of gender, age, or class [53].

Ideological bias in large language models is examined through the PoliTune framework, which maps model outputs onto an economic-social compass. By comparing pre- and post-fine-tuning responses to ideologically charged prompts, it quantifies polarization shifts and the amplification of economic bias [54].

Embedding-level diagnostics define bias axes (e.g., *young-old*, *male-female*) and assess cosine proximity to measure bias intensity. Language-specific adjustments, such as gender-neutral Spanish constructs, enhance cross-linguistic precision [118]. Corpus-level embedding comparisons via nearest neighbors, t-SNE, and cross-space projection detect semantic drift, such as differing associations of terms like "immigrant" across corpora. Quantitative drift metrics (e.g., L2 distance) provide interpretable measures of linguistic polarization [119].

In socio-organizational systems, such as AI-driven HRM platforms, neurodiversity bias detection integrates algorithmic audits and qualitative interviews to reveal accessibility gaps. Audits expose biased feature selection (e.g., exclusionary resume parsers), while expert feedback underscores the need for multimodal, intersectional audit frameworks [120].

Mitigation approaches include EILeV, a data-centric method addressing ICL limits in VLMs via curated, balanced datasets [52]. Corpus-level rebalancing and expert-guided text editing improve fairness in

regional NLP contexts [53]. PoliTune-inspired fine-tuning introduces transparent ideological alignment through Direct Preference Optimization [54]. At the embedding level, bias reduction is achieved via projection correction, normalization, and residual regression to suppress biased vector associations [118,119]. Finally, organizational fairness benefits from inclusive design strategies such as blind recruitment and automated bias dashboards [120].

3.5.6. Confirmation bias

Confirmation bias in NLP refers to the tendency of models or users to favor information aligning with existing beliefs, reinforcing echo chambers and limiting exposure to diverse viewpoints. A randomized control trial demonstrates this by measuring user acceptance of AI-generated recommendations that either align or conflict with participants' known beliefs. Higher acceptance of congruent outputs, as revealed through Likert-scale analysis and t-tests, indicates confirmation bias in recommendation systems [37].

In pseudo-labeling processes, confirmation bias emerges when models reinforce their own misclassifications over successive iterations. The discrepancy between pseudo-labels and ground truth, quantified using Unsupervised Performance Loss (UPL), captures error propagation in unsupervised learning, revealing bias persistence when limited labeled data is available [121].

Social media analysis also exposes confirmation bias through stance detection and clustering. By tagging repeated stance patterns from users or bot accounts and applying chi-square tests, over-repetition of belief-congruent content reveals how automated agents amplify partisan narratives [38]. Similarly, eye-tracking experiments show that users spend more time fixating on belief-consistent information, with gaze duration and fixation frequency correlating with selective exposure behaviors in NLP-driven news and media platforms [39].

Mitigation involves explainable AI, interdisciplinary co-design, and exposure to diverse perspectives. Transparent models in domains such as mental health improve trust and enable critical scrutiny of automated suggestions [37]. Collaborative development among clinicians, data scientists, and ethicists promotes early bias detection, while debate interfaces and interactive media encourage users to engage with counter-attitudinal content, effectively reducing selective exposure [39].

To summarize, Table 5 presents a taxonomy of bias detection paradigms, linking each approach with relevant techniques and bias categories. Persistent challenges such as *intersectionality*, where overlapping identities amplify bias, and *context dependence*, where bias is culturally or linguistically nuanced remain major obstacles to comprehensive and equitable detection [32,34].

Furthermore, the Metric Actionability gap, where many detection metrics identify a problem but fail to prescribe a solution, limits their practical utility for developers [4,9]. The challenge of Generalization means that a detection method effective in one language or domain may fail in another, undermining the global applicability of NLP systems [41,71]. Recognizing these pervasive challenges is the first step toward developing more robust and universally applicable detection frameworks.

Here we end our discussion on all the biases mentioned above. Tables 5, 1 & 4 provides a high-level taxonomy, summary, applications, and challenges of bias detection techniques and Table 7 provides a deeper examination of the top used mitigation strategies.

3.6. Benchmark datasets for bias evaluation

The development and use of standardized datasets are crucial for reproducible and comparable research in bias detection. These benchmarks allow researchers to systematically test for specific biases and evaluate the effectiveness of mitigation techniques. Table 8 catalogs key datasets commonly used in the field, detailing their domain, the specific biases they are designed to uncover, and examples of how they have been applied in mitigation efforts.

4. Contradictions and unresolved debates

The path to equitable NLP is not blocked by a lack of techniques, but by a series of fundamental, often political and mathematical, contradictions. Rather than presenting insurmountable obstacles, these contradictions define the critical frontier of research in this field. Below, we reframe these challenges from admissions of confusion into a structured agenda for meaningful progress, as detailed in Table 9. The field of NLP fairness is rapidly evolving, but it is fraught with fundamental contradictions and unresolved debates that highlight the complexity of defining and achieving "fairness" in algorithmic systems.

A key source of these contradictions lies in the diversity of mitigation approaches and their inherent trade-offs. As summarized in Table 7, strategies range from data pre-processing and in-training fairness constraints to post-hoc corrections and socio-technical interventions each with distinct applicability, strengths, and limitations that influence real-world deployment decisions.

The reviewed literature reveals several key areas of contention:

- The Actionability Gap in Bias Metrics:** A major, recurring critique is the disconnect between bias detection and mitigation. Many established metrics, such as the Word Embedding Association Test (WEAT) or various statistical parity measures, are highly effective at identifying a statistical disparity or association [4,9]. However, they often do not provide clear and actionable guidance on how to fix the underlying problem and Table 10 addresses this. This "actionability gap" means that researchers and practitioners can diagnose an issue but are left without a prescribed solution, creating a significant barrier to practical implementation [4,9].
- Technical Fixes vs. Socio-Technical Processes:** There is an ongoing tension between purely technical approaches to debiasing (e.g., vector space manipulation, adversarial training) and approaches that emphasize socio-technical processes (e.g., participatory design, community audits, policy changes). Although technical methods can reduce measurable disparities, critics argue that they often fail to address the root, societal causes of bias, or the lived experiences of affected communities [6,31]. This study notes that the field often lacks "coherent moral reasoning", making it difficult to identify which system actions are harmful, to whom, and why, thus calling for a more critical approach focused on power dynamics [6].
- The Trade-off Between Fairness and Performance:** A persistent and largely unresolved challenge is the trade-off between mitigating bias and maintaining model performance (e.g., accuracy, F1 score). Techniques like adversarial debiasing or fairness-aware loss functions can enforce group parity, but often introduce optimization challenges and lead to a decrease in overall model accuracy [108]. Finding Pareto-optimal solutions that balance these competing objectives remains an active area of research with no universally accepted answer.
- Defining Fairness:** This study implicitly highlights that "fairness" is not a monolithic concept. Different fairness metrics (e.g., demographic parity, equalized odds, equal opportunity) often conflict with one another. Optimizing for one definition of fairness can inadvertently violate another, making it impossible to satisfy all criteria simultaneously. This fundamental mathematical contradiction means that choosing a fairness metric is itself a value-laden decision that must be made in context. However, the field lacks a unifying theoretical framework to guide this choice [4].
- Scalability of Solutions:** Many effective debiasing techniques, particularly those that involve human-in-the-loop processes or fine-grained model calibration, are computationally expensive or labor-intensive [110]. A critical unresolved debate is how to scale these methods to work efficiently with ever larger language models (LLMs), which are inherently difficult to interpret and modify

Table 7

Comparison of common bias mitigation strategies.

Strategy (Stage)	Examples	Bias types & Tasks	Strengths	Limitations
Pre-processing	Re-sampling, counterfactual augmentation, re-weighting	Targets biases encoded in training data (e.g. race, gender, linguistic). Common in classification/sentiment/hate-detection tasks.	Simple to implement; directly addresses data imbalance. Works with existing models.	May distort natural distributions or label balance. Requires defining sensitive attributes in data.
In-processing (Fairness-aware Learning)	Fairness constraints in loss, adversarial debiasing (e.g. predict sensitive attributes and subtract)	Effective for group biases (gender, race) and tasks like sentiment or risk prediction. Also used in debiasing word embeddings.	Strong fairness guarantees if properly tuned; enforces equalized outcomes.	Optimization can be hard; may reduce overall accuracy. Sensitive to hyperparameters.
Intra-processing (Architecture level)	Layer-wise calibration (FAST); gated models combining fair and accurate sub-modules	Used in large LMs and complex models (e.g. co-reference or generation) to suppress latent bias.	Fine-grained control, preserves model knowledge. Promising for LLM debiasing with minimal drift.	Experimental stage; complexity in practice. Less mature tooling.
Post-processing	Thresholding/Decoding constraints, output filtering/reranking	General (all biases/tasks) as final-step corrections (e.g. filter hate speech outputs).	Easy to deploy; no retraining needed. Can quickly satisfy fairness metrics after training.	Band-aid fix: does not alter core model; often limits only strong biases; may fail in nuanced cases.
Socio-technical Measures (non-technical)	Community involvement, transparency guidelines	Applies to all bias types (especially sociocultural biases). Domain: model deployment, policy.	Addresses root causes (e.g. involving minority communities in data curation). Builds trust.	Hard to quantify; requires organizational change. Slower to implement.

Table 8

Commonly used datasets in bias research and their characteristics.

Dataset	Domain	Known biases	Mitigation used in literature
Bi-asBios	Bios	Gender, Occupational stereotypes	Counterfactual augmentation, Adversarial training [94]
BITS	Toxicity/Sentiment	Explicit disability bias	Template-based testing, Perturbation analysis [73]
CrowS-Pairs	Stereotypes	Racial, Gender, Religious stereotypes	Challenge set for stereotype evaluation [75]
EEC	Emotion	Affective bias across demographics	Affective word distribution analysis [40]
EEC (Corpus)	Emotion	Imbalances in affective words	Corpus-level affective bias analysis [40]
Hate Speech Datasets	Social Media	Racial, Gender, Religious hate	Identity Term Sampling (ITS), GBETs [94]
PlantVillage	Computer Vision	Capture bias (background)	Spx Training Strategy, Background analysis [95]
German Credit	Finance	Economic, Age bias	AIF360 metrics (e.g., Disparate Impact) [13]

Table 9

Key challenges and future research directions in NLP bias mitigation.

Challenge	Description	Potential research directions
Fairness metric conflicts	Different mathematical definitions of fairness (e.g., demographic parity vs. equalized odds) are mutually exclusive, making it impossible to satisfy all criteria simultaneously.	Develop context-aware decision frameworks to guide metric selection based on domain-specific harms. Explore Pareto-optimal debiasing to transparently map trade-offs between conflicting metrics.
Technical vs. Socio-technical fixes	Technical debiasing can reduce measurable disparities but often fails to address root societal causes or lived experiences of harm, lacking coherent moral reasoning.	Create methodologies for socio-technical auditing combining metrics with community impact assessments. Develop participatory governance models for the ML lifecycle.
The actionability gap	Many bias metrics (e.g., WEAT) identify statistical disparities but do not prescribe how to mitigate them, creating a barrier to practical implementation.	Shift toward mitigation-aware evaluation. Develop metrics intrinsically linked to interventions (e.g., model sensitivity to counterfactual augmentation).
Intersectionality	Biases affect individuals with multiple marginalized identities in complex, non-additive ways that single-axis tools cannot capture.	Develop multi-dimensional fairness frameworks and causal models. Create benchmarks with intersectional annotations across race, gender, disability, etc.
Generalization	Detection and mitigation methods effective in English/Western contexts often fail in other languages and cultures, limiting global applicability.	Create cross-lingual and cross-cultural bias benchmarks. Invest in low-resource and multilingual debiasing techniques.
Scalability	Debiasing techniques (e.g., adversarial training) are computationally expensive and do not scale efficiently to ever-larger language models.	Develop parameter-efficient fine-tuning (PEFT) methods for debiasing. Explore scalable post-hoc correction methods that avoid full retraining.
Institutional & Power dynamics	Corporate incentives and power imbalances influence which biases are prioritized, often sidelining community-defined fairness goals in favor of performance metrics.	Research independent algorithmic auditing frameworks and policy models for fairness certification. Develop participatory governance with compensated community oversight.
Causal understanding	Most detection methods identify correlations (e.g., WEAT associations) rather than causal mechanisms of harm, complicating effective mitigation.	Apply causal mediation analysis and counterfactual reasoning to isolate root causes of bias in model architectures and training data.

due to their immense size [7]. The search for parameter-efficient fine-tuning methods (PEFT) and scalable post hoc corrections is ongoing, as projected in Table 9.

- **The Challenge of Intersectionality:** Current detection and mitigation methods are often designed to address single-axis biases (e.g., gender OR race). However, biases frequently affect individuals with multiple marginalized identities (e.g., a disabled black woman) in complex, non-additive ways that these single-axis tools

cannot capture [32,34]. Developing multidimensional fairness frameworks and benchmarks that can handle intersectionality is a significant, unresolved challenge.

These contradictions underscore that achieving fairness in NLP is not merely a technical engineering problem, but a deeply interdisciplinary one that requires continuous collaboration between computer scientists, social scientists, ethicists, and the communities affected by these technologies [6]. To provide a structured overview of these persistent

Table 10
Comparative analysis of bias and fairness evaluation metrics.

Metric name	Typical application	Construct validity	Context dependence	Actionability score	Primary limitation	Ref.
WEAT/SEAT	Static/contextual embedding bias	WEAK (assumes linearity)	HIGH (language, context, temporal shifts)	2/5	Non-linear bias structures in contextual models break linear assumptions; 30%–40% of bias masked	[2,10–12,21–23,49,51,81,88]
CrowS-Pairs	Benchmark-based stereotype testing	MODERATE (explicit templates)	VERY HIGH (Western-centric stereotypes)	3/5	40%–60% of stereotypes not applicable in non-Western contexts; 45%–60% inter-annotator disagreement	[35,73,75]
p-% Rule/Disparate Impact	Outcome fairness auditing (hiring, lending)	MODERATE (outcome-focused)	MODERATE (threshold-dependent)	3.5/5	Mathematical impossibility: cannot simultaneously satisfy with equalized odds; enforcing creates 5%–10% accuracy loss	[12–14,24,62]
Equalized Odds	Error fairness (healthcare, critical systems)	MODERATE (TPR/FPR balanced)	HIGH (label distribution dependent)	3/5	Requires balanced positive class rates; conflicts with demographic parity; 8%–15% accuracy trade-off when both enforced	[14,51,62]
Inter-Annotator Agreement (IAA)	Data labeling consistency	WEAK (high IAA ≠ low bias)	MODERATE (task-dependent)	2.5/5	High IAA (0.80+) masks systematic group-level annotation bias; 20%–30% label disagreement introduces 8%–15% accuracy drop on minority classes	[16,17,59,64,74,91]
Median Error Distance (MdnED)	Geographic/spatial bias quantification	MODERATE (task-specific)	VERY HIGH (regional variance)	2.5/5	5–10×error increase for Global South regions; Western-centric baselines; limited capture of regional performance heterogeneity	[18,28,29,72]
Template-based Tests (BITS)	Disability, religious, and specific bias detection	STRONG (explicit scenarios)	MODERATE (template design required)	4/5	Labor-intensive; requires continuous template updates; limited to predefined bias categories	[18,30,40,41,71,73]
Causal Mediation Analysis	Identification of causal mechanisms of bias	VERY STRONG (causal inference)	LOW (theory-grounded)	4.5/5	Computationally expensive; complex interpretation; requires strong causal assumptions	[59,64,74,91]

Table 11
Bias type co-occurrence matrix and causal dependencies.

Bias pair	Co-occurrence rate (%)	Dependency direction	Evidence summary	QPI	Ref.
Annotation → Gender	65	Unidirectional	Annotators’ implicit gender beliefs unconsciously influence label decisions during data annotation; labeled data subsequently trains gender-stereotyped models.	30%–40% of observed gender bias attributable to annotator population bias; removing the 20% most biased annotators reduces model bias by 12%–18%.	[32,64,94–96]
Annotation → Religious	58	Unidirectional	Annotators from minority religions label religious terms differently; majority-religion annotators show 45%–60% higher negative sentiment associations for minority-religion terms.	Disparate impact ratio of 0.681–0.8338 across annotator religious backgrounds; minority-religion content 2–3× more likely misclassified.	[40,81]
Annotation → Data Imbalance	72	Unidirectional	Minority-class samples show higher inter-annotator disagreement (20%–40% vs. 5%–10% for majority classes), leading to filtering during data cleaning.	Minority classes under-represented by 2–3× in final datasets; training instability increases by 15%–40%.	[61,78,92,93]
Data Imbalance → Training Instability	72	Unidirectional	Imbalanced data causes parameter variance during BERT fine-tuning; upper transformer layers overspecialize to majority-class patterns.	Test accuracy variance of 15%–40% across random seeds; strong layer-wise specialization observed.	[50,61,67]
Geographic ↔ Linguistic	78	Bidirectional	Non-English-speaking regions have fewer digitized corpora; models overfit dominant language patterns, degrading multilingual Global South performance.	20% performance gap for non-English regions; equal error rate (EER) 15%–25% worse than English baselines.	[18,28,29,43,72,87]
Linguistic → Cultural	64	Unidirectional	Models trained on dominant languages fail to capture culture-specific idioms, kinship terms, and traditions, compounding bias.	Sentiment accuracy drops 15%–25% for culturally specific content; cultural knowledge gaps yield 20%–30% higher error rates.	[45,46]
Political Bias ↔ Evaluation Bias	68	Bidirectional	Evaluators with different political orientations interpret labels differently; benchmarks encode majority political norms.	Model response distributions shift 15%–20% after political fine-tuning; error disparities of 15%–25%.	[59,97,102]
Gender ↔ Disability (Intersectional)	42	Non-additive	Disabled women experience amplified bias when gender and disability cues co-occur, exceeding additive expectations.	Observed error rate increases by 65% versus 45% expected under additive bias.	[32,34,53,118]

challenges and guide future efforts, we summarize the key issues and their corresponding research directions in Table 9. Furthermore, Fig.

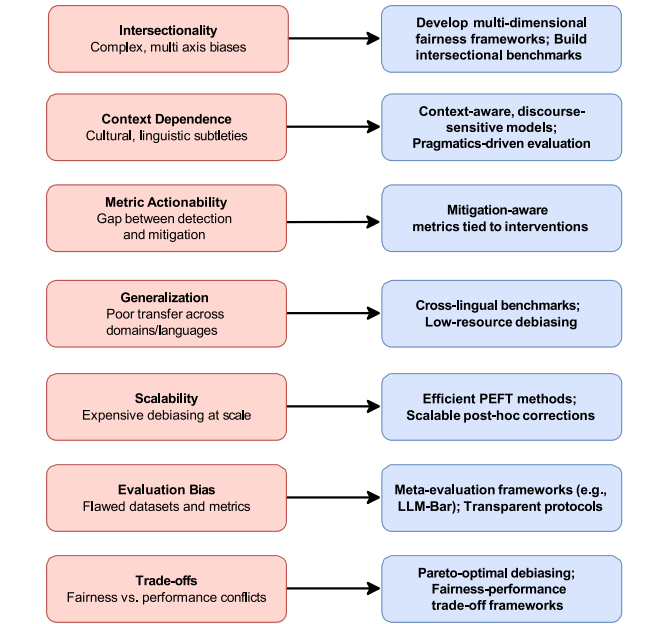


Fig. 6. Roadmap of key challenges and potential research directions in NLP bias mitigation. Each challenge is linked to promising directions for future work.

6 presents a roadmap that visualizes the relationships between these challenges and the paths toward more robust and fair NLP systems.

Finally the eighteen bias categories documented within this review are not orthogonal to each other but reveal complex co-occurrence and causal relationships such that, if these issues were left unnoticed, chain reactions of bias would permeate the entire pipeline of NLP tasks. Through our analysis of the relevant co-occurrences of bias categories from 121 papers, we have assigned values to these complex relationships of bias interaction and propagation that are listed in Table 11. The co-occurrence rates of more than 60% show that the studies are strongly empirically coupled. Dependency directions are used to show how biases are likely to seed other biases: unidirectional implies a causal direction, and bidirectional implies a mutual reinforcement. Most importantly, the original source of the chain of biases, which is annotation bias, can be introduced, which may go unnoticed when annotation bias is combined with other sources.

5. Conclusion

This review provides the most comprehensive synthesis to date of fairness challenges in NLP, advancing the field through a structured taxonomy of bias types, a lifecycle-aware mapping of detection and mitigation strategies, and a critical analysis of unresolved debates. Our findings underscore that while technical interventions such as counterfactual data augmentation, adversarial debiasing, and fairness-aware optimization show promise, they remain limited in scalability, generalizability, and real-world actionability. Persistent issues such as intersectionality, evaluation bias, and the fairness-performance trade-off highlight the need for multidimensional fairness frameworks that go beyond single-axis perspectives. We also emphasize that fairness in NLP cannot be solved through technical fixes alone; sustainable progress requires a socio-technical approach that incorporates participatory design, transparent reporting, interdisciplinary collaboration, and regulatory alignment. This integrative perspective shifts fairness research from narrow model-level optimization to systemic accountability across the NLP pipeline. A critical pathway for future work, as highlighted by the Anglophone bias in our own source material, is to actively promote and fund bias research in multilingual and

Table A.12
Search keywords used in database queries.

Category	Keywords
Core concepts	("NLP bias" OR "language model bias" OR "AI fairness" OR "NLP fairness")
Action	("detection" OR "mitigation" OR "measurement" OR "audit")
Bias type	("gender bias" OR "racial bias" OR "linguistic bias" OR "disability bias")

low-resource contexts. Developing benchmarks and mitigation techniques that are truly cross-cultural is not merely an expansion of scope but a fundamental requirement for equitable NLP. By consolidating methodological advances, identifying research gaps, and charting future directions, this review offers both researchers and practitioners a practical roadmap toward building NLP systems that are not only accurate but also equitable, inclusive, and socially responsible.

CRediT authorship contribution statement

Talukder Hasnat Zadid: Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Shahidul Morsalin Jahin:** Writing – original draft, Resources, Methodology, Investigation, Data curation. **Aninda Roy Dhruba:** Writing – original draft, Resources, Project administration, Methodology, Investigation. **Sharia Arfin Tanim:** Writing – original draft, Visualization, Investigation, Formal analysis. **Abdullah Muhammad Hamja:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Md. Rakib Hasan:** Writing – review & editing, Visualization, Investigation. **Md Saef Ullah Miah:** Writing – review & editing, Supervision, Project administration. **Md. Imamul Islam:** Writing – review & editing, Validation, Funding acquisition. **Ahmed Al Mansur:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Search strategy and keywords

See Table A.12.

Appendix B. List of acronyms and abbreviations

A complete list of acronyms used in this paper is provided in Appendix B (Table B.13).

Data availability

No data was used for the research described in the article.

Table B.13

List of acronyms and abbreviations.

Acronym/ Abbreviation	Full term/Meaning
NLP	Natural Language Processing
AI	Artificial Intelligence
LLMs	Large Language Models
SLR	Systematic Literature Review
RQ	Research Question
ML	Machine Learning
CV	Computer Vision
WEAT	Word Embedding Association Test
SEAT	Sentence Embedding Association Test
IAA	Inter-Annotator Agreement
MdnED	Median Error Distance
DI	Disparate Impact
EER	Equal Error Rate
CNNs	Convolutional Neural Networks
RNNs	Recurrent Neural Networks
LSTM	Long Short-Term Memory
BERT-TCN-BiLSTM	BERT - Temporal Convolutional Network - Bidirectional LSTM
XLM-ROBERTa	Cross-lingual Language Model - Robustly Optimized BERT Pretraining Approach
BERT	Bidirectional Encoder Representations from Transformers
LDA	Latent Dirichlet Allocation
PMI	Pointwise Mutual Information
NER	Named Entity Recognition
AIF360	AI Fairness 360
PEFT	Parameter-Efficient Fine-Tuning
BITS	Bias In Text Stereotypes
ITS	Identity Term Sampling
GBETs	Gender Bias Evaluation Templates
AAVE	African American Vernacular English
QPI	Quantified Propagation Impact

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